Question format for ERS Lab Test

Electromagnetics & Radiating Systems Laboratory

(22EC52) - An Integrated Course

Expt. No.	Experiments
1.	The open surfaces ρ = 2.0 m and ρ = 4.0 m, *z* = 3.0 m and *z* = 5.0 m, and ϕ = 20° and ϕ = 60° identify a closed surface. Find a) the enclosed volume, b) the total area of the enclosed surface. Write a MATLAB program to verify your answers.
2.	An infinite uniform linear charge ρ_L = 2.0 nC/m lies along the x-axis in free space, while point charges of 8.0 nC each are located at (0,0,1) and (0,0,-1). Find Electric field at (2,3,4) using analytical method. Verify your answer using a MATLAB program.
3.	A point charge of 1.0 C is located at (0,0,1). Find analytically the total electric flux going through the infinite xy plane located at origin. Verify your answer using a MATLAB program.
4.	Find curl of vector -y $\hat{\imath}$ + x $\hat{\jmath}$ + 0 $\hat{k}$ using analytical method. Verify your answer using a MATLAB program.
5.	Design and analyze the performance of a Dipole Antenna for ____GHz frequency using Ansys HFSS. Use radius of dipole as 1mm. Plot S11, VSWR and polar plot of gain.
6.	Design and analyze the performance of a Monopole antenna for _____MHz frequency using Ansys HFSS. Use radius of monopole as 1cm. Plot S11, VSWR and polar plot of gain.
7.	Design and analyze the performance of coaxial probe fed rectangular microstrip patch antenna for 2.4 GHz frequency using Ansys HFSS. Plot S11, VSWR and polar plot of gain.
8.	Design and analyze the performance of inset-fed rectangular microstrip patch antenna for 2.4 GHz frequency using Ansys HFSS. Plot S11, VSWR and polar plot of gain.
9.	Design and analyze the performance of inset-fed Microstrip patch antenna array for 2.4GHz frequency using Ansys HFSS. Plot S11, VSWR and polar plot of gain.
10.	Design and analyze the performance of Yagi-Uda antenna for 0.9 GHz frequency using Ansys HFSS. Consider radius of reflector, driven element and director as 1mm. Plot S11, VSWR and polar plot of gain.
11.	Design and analyze the performance of Helical antenna for 2GHz frequency using Ansys HFSS. Plot S11, VSWR and polar plot of gain.

Note:-

Preliminary data will be given for experiments 7,8,9 and 11.

Coaxial probe fed rectangular microstrip patch antenna for 2.4 GHz

Preliminary data :-

Draw radiating patch with dimensions as per the design.

Take the substrate FR4 having relative permittivity of 4.4 and thickness 1.6 mm.
Consider the size of the substrate as 70mm, 70mm, 1.6mm on X Y Z axis.

Consider the size of ground plane as 70mm, 70mm on X Y axis.

Make a cut in ground by using a circle of 1.6mm radius positioned at
-5mm,0mm,0mm on X Y Z axis.

For **coaxial cable** use outer cylinder with radius 1.6mm and height -5 mm below ground plane positioned at -5mm,0mm,0mm on X Y Z axis. Use inner cylinder with radius 0.7mm and height -5 mm below ground plane positioned at -5mm,0mm,0mm on X Y Z axis. Use second inner cylinder with radius 0.7mm and height 1.6 mm above the ground plane positioned at -5mm,0mm,0mm on X Y Z axis for the inner cylinder that passes through the substrate.

For giving excitation to bottom of coaxial cable and thereby to patch use circle of radius 1.6mm positioned at -5mm, 0mm, -5mm on X Y Z axis. Use terminal wave port as excitation.

Create radiation box manually with material as air with size 70mm, 70mm, 20mm on X Y Z axis.
Select only 5 faces, 4 side faces and top face while assigning radiation boundary.

Create infinite sphere with Phi 0-360 and Theta 0-180 for plotting polar gain plot.

Inset-fed rectangular microstrip patch antenna for 2.4 GHz

Preliminary data :-

Draw radiating patch with dimensions as per the design.

Take the substrate FR4 having relative permittivity of 4.4 and thickness 1.6 mm.
Consider the size of the substrate as 70mm, 70mm, 1.6mm on X Y Z axis.

Consider the size of ground plane as 70mm, 70mm on X Y axis.

For feedline use the size 35mm on X axis and 3mm on Y axis with position 0mm, -1.5mm, 1.6mm.
To make a cut in patch while connecting feedline to patch use rectangle dimensions as -9.5mm on X axis and 5mm on Y axis with position 14.7mm, -2.5mm, 1.6mm.

For excitation(feed) rectangle (in YZ plane) use size 3mm on Y axis, 1.6mm on Z axis with position 35mm, -1.5mm, 0mm on X Y Z axis.

Create the radiation box manually with material as air with size 80mm, 80mm, 40mm on X Y Z axis.

Create infinite sphere with Phi 0-360 and Theta 0-180 for plotting polar plot.

Inset-fed Microstrip patch antenna array for 2.4 GHz

Preliminary data :-

Draw radiating patch with dimensions as per the design.

Take the substrate FR4 having relative permittivity of 4.4 and thickness 1.6 mm.
Consider the size of the substrate as 70mm, 70mm, 1.6mm on X Y Z axis.

Consider the size of ground plane as 70mm, 70mm on X Y axis.

For feedline use the size 35mm on X axis and 3mm on Y axis with position 0mm, -1.5mm, 1.6mm.
To make a cut in patch while connecting feedline to patch use rectangle dimensions as -9.5mm on X axis and 5mm on Y axis with position 14.7mm, -2.5mm, 1.6mm.

For excitation(feed) rectangle (in YZ plane) use size 3mm on Y axis, 1.6mm on Z axis with position 35mm, -1.5mm, 0mm on X Y Z axis.

Create a copy of the antenna using duplicate along line option along positive Y axis.
Unit both substrates and both ground planes.

Create the radiation box manually with material as air with size 120mm, 190mm, 40mm on X Y Z axis having position -60mm, -60mm, -20mm on X Y Z.

Create infinite sphere with Phi 0-360 and Theta 0-180 for plotting polar plot.

Helical antenna for 2 GHz frequency

Preliminary data :-

For drawing reference circle i.e. cross section (XZ plane) of helix use dimensions center position as 1.6cm, 0cm, 0cm with radius 0.015cm.

Consider,
Center of helix – 0cm,0cm,0cm
Direction of helix – 0cm,0cm,7.66cm
Pitch (Longitudinal spacing) – 2cm
Number of turns – 3.83

For circular ground plane consider radius as 3.75 cm and center position as 0cm,0cm,-2cm on X Y Z axis.

Use feed rectangle (XZ plane) with size 0.03cm on X axis ,2cm on Z axis with position 1.585cm, 0cm, -2cm on X Y Z axis.

For radiation boundary cylinder use dimensions 5cm as radius, 16cm as height with 0cm, 0cm, -3cm as position.

Create infinite sphere with Phi 0-360 and Theta 0-180 for plotting polar plot.

Change mesh setting to course to simulate helix structure.